Week 7 - Digital Signatures

MAT260: Cryptology

Gary Hobson

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Introduction:

The assigned computer problems this week are from Chapter 13, Problem 2.

Make sure to run your code so all relevant computations/results are displayed and then export your work as a PDF file for submission.

Chapter 13 Problems:

Problem 2:

To solve the problem, I used the provided RSA parameters for Alice and Bob. First, I computed Bob's private decryption key dB by calculating the modular inverse of his public exponent eB modulo ϕ(nB), where:

ϕ(nB) = (pB − 1)(qB − 1).

I then used dB to decrypt the received encrypted message m1 and signature s1. After obtaining the original message m and signature s, I verified the signature by checking whether:

seA mod nA = m.

Since the verification passed, I concluded that the message is authentic and was indeed signed by Alice.

```
format long g
% Public and private key components
nA = sym('171024704183616109700818066925197841516671277');
```

```
nB = sym('8390735427343693592608713559390626227476331O9');
eA = sym('1571');
eB = sym('87697');
pB = sym('98763457697834568934613');
qB = sym('84957894578934573457 93');
```

```
% Encrypted values received by Bob
m1 = sym('41872655399709425857798005506130515094054 7956');
s1 = sym('74914264964154810152013363473686575288327 7237');

% Step 1: Compute Bob's private key dB
phiB = (pB - 1)*(qB - 1);
dB = modInverse(eB, phiB);

% Step 2: Decrypt m and s using dB
m = powermod(m1, dB, nB);
s = powermod(s1, dB, nB);

% Step 3: Verify Alice's signature
check = powermod(s, eA, nA);

% Display results
fprintf('Decrypted message m = %s\n', char(m));
```

```
Decrypted message m = 19012507151504022505
```

```
fprintf('Decrypted signature s = %s\n', char(s));
```

```
Decrypted signature s = 150270996499036309478023705411245214416829627
```

```
fprintf('Verification result (s^eA mod nA) = %s\n', char(check));
```

```
Verification result (s^eA mod nA) = 19012507151504022505
```

```
if check == m
    fprintf('Signature is VALID and message is from Alice.\n');
else
    fprintf('Signature verification FAILED.\n');
end
```

```
Signature is VALID and message is from Alice.
```

```
% --- Helper function ---
function inv = modInverse(a, m)
    [g, x, ~] = gcd(a, m);
    if g ~= 1
        error('Inverse does not exist');
    else
        inv = mod(x, m);
    end
end
```